

Protocol Stack

Module 05

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Video: <https://www.youtube.com/watch?v=KanyeVKL6cs>

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Dr Navdeep Singh, Department of Computer Engineering, Punjabi University, Patiala

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0:00 [Music]

0:15 my dear students hello I welcome you all in our series of lectures on cyber security today we are going to study

0:24 about protocol stack after going through this lesson you will be able to understand the needs

0:31 of protocols and standards also you will be able to identify various protocols at different

0:38 layers of protocol stack so let us begin with it the protocol stack is an implementation of a

0:47 computer networking protocol suit the terms are often used interchangeably the suit is the

0:54 definition of the protocols and the stack is the software implementation of them a protocol stack is a complete set of

1:03 network protocol layers that work together to provide networking capabilities it is called aack because

1:12 it is typically designed as a hierarchy of layers each supporting the one above it and using those below

1:20 it individual protocols within a suit are often designed with a single purpose in mind this modelization makes design

1:29 and EV valuation easier because each protocol module usually communicates with two others

1:37 they're commonly imagined as layers in a stack of protocols the lowest protocol always

1:44 deals with low level physical interaction of the hardware every higher layer adds more

1:52 features user applications usually deal only with the topmost layers in Practical implementation

2:01 2 minutes, 1 second protocol Stacks are often divided into three major sections media transport and applications a particular operating

2:10 system or platform will often have two well-defined software interfaces one
between the media and transport layers and one between the
2:19 transport layers and applications the media to transport interface defines how
transport protocol
2:26 software makes use of particular media and Hardware types for example this
interface level would Define how TCP IP
2:36 transport software would talk to the ethernet Hardware the application to
transport interface defines how application
2:45 programs make use of the transport layers for example this interface level would
Define how a web browser program
2:52 would talk to the TCP IP transport software next is OSI model
3:01 3 minutes, 1 second Osa stands for Open System interconnection model it is a
conceptual model that characterizes and standardizes the communication functions
3:10 of a telecommunication or Computing system without regard to their underlying
internal structure and Technology its goal is the
3:19 interoperability of diverse communication systems with the standard protocols the
model partitions a
3:26 communication system into abstraction layers the original version of the model
defines seven
3:34 layers the model is a product of the open systems interconnection project at the
International Organization for standardization maintained by
3:42 identification ISO 7498 the Osa model has seven
3:49 layers the principles that were applied to arrive at the seven layers can be briefly
summarized as number one a layer
3:58 should be created where a different abstraction is needed number two each layer
should perform a well- defined
4:08 function number three the function of each layer should be chosen with an eye
toward defining internationally standardized
4:16 protocols and number four the layer boundaries should be chosen to minimize the
information flow across the
4:24 interfaces number five the number of layers should be large enough that distinct
functions need not be thrown together in the same layer out of
4:33 necessity and small enough that the architecture does not become unwieldy figure
shows the OSI reference
4:42 model see this next figure the figure shows the exchange of data using USA
4:49 model D7 means the data unit at L 7 D6 means the data unit at layer six and so
4:58 on the process starts at layer 7 called the application layer and then moves from
layer to layer in descending
5:05 sequential order at each layer a header or possibly a trailer can be added to the
data

5:14 unit commonly the trailer is added only at Layer Two when the formatted data unit passes

5:22 through the physical layer it is changed into an electromagnetic signal and transported along a physical link

5:31 upon reaching its destination the signal passes into layer one and is transformed back into digital

5:38 form the data units then move back up through the OSI layers as each block of data reaches the

5:47 next higher layer the headers and trailers attached to it at the corresponding sending layer are removed

5:54 and actions appropriate to that layer are taken by the time it reaches L 7 the

6:01 6 minutes, 1 second message is again in a form appropriate to the application and is made available to the

6:08 recipient next is layers in the Osa model the various layers of Osa model

6:16 are number one physical layer in the seven layer USA model of computer networking the physical layer

6:24 or layer one is the first and the lowest layer the physical layer coordinates the functions required to carry a bitstream

6:33 over a physical medium it deals with a mechanical and electrical specifications of the interface and transmission

6:42 medium it also defines the procedures and functions that physical devices and interfaces have to perform for

6:50 transmission to occur now what are the services provided by physical layer in digital Communications symbol

6:58 rate also known as board rate and modulation rate is the number of symbol changes waveform changes or signaling

7:05 events across the transmission medium per time unit using a digitally modulated signal or a line

7:13 code the symbol rate is measured in board or symbols per second in the case of a line code the

7:20 symbol rate is the pulse rate in pulses per second number two providing a standardized interface to physical

7:29 transmission media including mechanical specifications of electrical connectors and cables for example maximum cable

7:36 length providing electrical specifications of transmission line signal level and impedance providing

7:44 radio interface including electromagnetic spectrum frequency allocation and specification of signal strength analog bandwidth etra and

7:53 providing specifications for IR over optical fiber or a wireless IR communication link

8:01 8 minutes, 1 second number three modulation in electronics and
Telecommunications modulation is the process of wearing one or more
8:09 properties of a periodic waveform called the carrier signal with a modulating signal
that typically contains information to be
8:18 transmitted a modulator is a device that performs modulation a demodulator is a
device
8:25 that performs demodulation the inverse of modulation a botom can perform both
8:32 operations number four line coding line coding consists of representing the digital
signal to be
8:40 transported by a waveform that is optimally tuned for the specific properties of the
physical Channel and of the receiving
8:49 equipment the pattern of voltage current or Photon used to represent the digital
data on transmission link is called the
8:57 line encoding the common types of line encoding are unipolar polar bipolar and
Manchester
9:07 encoding after line coding the signal is put through a physical Channel either a
transmission medium or a data storage
9:15 medium number five bit synchronization the sender and receiver not only must use
the same bit rate but
9:24 also must be synchronized at the bit level in other words the and the receiver
clocks must be
9:33 synchronized number six line configuration the physical layer is concerned with the
connection of devices
9:40 to the media in a point to-point configuration two devices are connected through a
dedicated Link in a multipoint configuration a
9:49 link is shared among several devices number seven physical
9:56 topology the physical topology defines how devices are connected to make a
10:02 network and number eight is transmission mode transmission mode defines the
10:09 direction of transmission between two devices Simplex half duplex or full duplex
next is data link
10:18 layer the data link cayer or Layer Two is the second layer of the seven layer USA
model of computer
10:26 networking this layer is the protocol layer that trans transfers data between
adjacent Network nodes in a wide area
10:32 network or between nodes on the same local area network segment the data link
clear provides the
10:40 functional and procedural means to transfer data between Network entities and
might provide the means to detect
10:47 and possibly correct errors that may occur in the physical layer now what are the
services provided by data link

10:55 CER number one framing a frame is the protocol data unit at the data link
11:03 cayer frames are the result of the final layer of encapsulation before the data is
transmitted over the physical
11:11 layer a frame is a unit of transmission in a link layer protocol and consist of a link
clear header followed by a
11:19 packet number two physical addressing if frames are to be distributed to different
systems on the
11:27 network the data link clear adds a header to the frame to define the sender or
receiver of the
11:34 frame if the frame is intended for a system outside the senders Network the
receiver address is the address of the
11:41 device that connects the network to the next one number three flow control flow
11:49 control is the process of managing the rate of data transmission between two nodes
to prevent a fast sender from
11:56 overwhelming a slow receiver it provides a mechanism for the receiver to control
the transmission speed so
12:04 that the receiving node is not overwhelmed with data from the transmitting node
number four error
12:11 control it allows the receiver to inform the sender if a frame is lost or damaged
during transmission and coordinates the
12:19 retransmission of those frames by the sender error control in the data link clear is
based on automatic repeat
12:27 request whenever an is detected specified frames are retransmitted number five
logical link control logical
12:37 link control refers to the functions required for the establishment and control of
logical links between local devices on a
12:45 network as mentioned this is usually considered a dll sublayer it provides services
to the network layer above it
12:53 and hides the rest of the details of the data link clear to allow different
Technologies to work seamlessly with the higher
13:01 13 minutes, 1 second layers most local area networking Technologies use the i e
82.2 LLC
13:08 protocol and number six is Media Access Control this refers to the procedures used
by devices to control access to the
13:17 network medium since many networks use a shared medium such as a single
network cable or a series of cables that are electrically
13:26 connected into single virtual medium it is necessary to have rules for managing the
medium to avoid
13:33 conflicts for example ethernet uses the csma CD method of media Access Control

- 13:40 while token ring uses token passing next is Network layer in the seven layer Osa model of computer
- 13:48 networking the network layer is layer three the network layer is responsible for packet forwarding including routing
- 13:57 through intermediate routers since it knows the address of neighboring Network nodes and it also manages quality of
- 14:05 service the services provided by Network CER are number one logical addressing every device that communicates over a
- 14:14 network has associated with it a logical address sometimes called a layer three address for example on the internet the
- 14:22 Internet Protocol is the network clear protocol and every machine has an IP address number two
- 14:30 routing moving data across a series of interconnected networks is probably the defining function of the network
- 14:37 layer it is the job of the devices and software routines that functions at the network layer to handle incoming Packers
- 14:45 from various sources and determine their final destination number three datagram
- 14:53 encapsulation the network layer normally encapsulate messages received from higher layers by placing them into
- 15:00 datagrams also called Packers with a network clear header number four fragmentation and
- 15:08 reassembly the network layer must send messages down to the data link layer for transmission some data link clear
- 15:16 technologies have limits on the length of any message that can be sent if the packet that the network CER
- 15:23 wants to send is too large the network cayer must split the packet up send each piece to the data link clear and then
- 15:30 have pieces reassembled once they arrive at the network clear on the destination machine number five error handling and
- 15:40 Diagnostics special Protocols are used at the network clear to allow devices that are logically connected or that are
- 15:47 trying to root traffic to exchange information about the status of host on the network or their devices
- 15:55 themselves various protocols at Network C are I C MP internet control message protocol igmp internet group management
- 16:04 protocol IP SEC Internet Protocol security IP version 4 or IP version 6 that is Internet
- 16:12 Protocol next is transport layer the transport layer is called the host to host transport layer in the

16:19 tcpip model the data communicated by the transport layer is encapsulated in a
transport layer pdu and sent in a
16:28 network layer the network clear nodes transfer the transport pdu intact without
decoding or
16:35 modifying the content of the pdu in this way only the peer transport entities
actually communicate using the
16:43 pdus of the transport protocol it provides services such as connection oriented data
stream
16:50 reliability flow control and multiplexing let us discuss them one by
16:56 one number one connection oriented communication it is normally easier for an
application to interpret a connection
17:04 as a data stream rather than having to deal with the underlying connectionless
models such as the datagram model of the
17:11 user datagram protocol and of the Internet Protocol number two same order
17:19 delivery the network CER doesn't generally guarantee that packet of data will
arrive in the same order that they
17:26 were sent but often this is a desirable feature this is usually done through the use
of segment numbering with the
17:34 receiver passing them to the application in order number three reliability Packers
may be lost during
17:43 the transport due to network congestion and errors by means of an error detection
code such as checksum the
17:50 transport protocol May check that the data is not corrupted and verify Crypt reip
by sending an act or neack message
17:58 to the sender automatic repeat request schemes may be used to retransmit lost or
corrupted
18:06 data number four flow control the rate of data transmission between two nodes
must sometimes be
18:14 managed to prevent a fast sender from transmitting more data then can be sported
by the receiving data buffer causing a buffer
18:23 overrun this can also be used to improve efficiency by reducing buffer underrun
18:30 number five congestion avoidance congestion control can control traffic entry into
a telecommunications
18:38 Network so as to avoid congestive Collapse by attempting to avoid over
subscription of any of the processing
18:45 all link capabilities of the intermediate nodes and networks and taking resource
reducing steps such as
18:52 reducing the rate of sending packets for example automatic repeat request may
keep the n work in a congested State

19:01 19 minutes, 1 second this situation can be avoided by adding congestion avoidance to the flow control including slow start this keeps the bandwidth

19:10 consumption at a low level in the beginning of the transmission or after packet retransmission and number six is

19:19 multiplexing ports can provide multiple end points on a single node for example the name on a postal address is a kind

19:26 of multiplexing and distinguish is between different recipients of the same location Computer Applications will each

19:35 listen for information on their own ports which enables the use of more than one network service at the same

19:42 time it is the part of the transport layer in the tcpip model but of the session layer in the Osa

19:50 model let us now learn about two most important protocols at transport layer number one transmission control

20:00 protocol TCP is a connection oriented protocol which means a connection is established and maintained until the

20:08 application programs at each end have finished exchanging messages it determines how to break

20:15 application data into packets that networks can deliver sends packet to and accepts packet from the network CER

20:23 manages flow control and because it is meant to provide error free data transmission handles retransmission of dropped or

20:31 garbed packets as well as acknowledgement of all packets that arrive and second is user datagram protocol the user datagram protocol is a

20:40 transport layer protocol defined for use with the IP network layer protocol it provides a best effort datagram service

20:46 to an end system the service provided by UDP is an unreliable service that provides no guarantees for delivery and

20:54 no protection for duplication the Simplicity of UDP reduces the overhead from using the protocol and the services

21:01 21 minutes, 1 second may be adequate in many cases UDP provides a minimal unreliable best effort message passing transport to

21:08 applications and upper layer protocols compared to other transport protocols UDP and its UDP light variant are unique

21:17 in that they do not establish endtoend connections between communicating and systems UDP communication consequently

21:24 does not incur connection establishment and tear down overheads and there is minimal Associated end System state

21:32 because of these characteristics UDP can offer a very efficient communication transport to some applications but has no inherent

21:40 congestion control or liability on many platforms applications can send UDP datagrams at the line rate of the link

21:48 interface which is often much greater than the available path capacity and doing so would contribute to congestion

21:55 along the path applications therefore need to be designed responsibly next is session layer the

22:04 session layer provides the mechanism for opening closing and managing a session between end user application

22:11 processes that is a semi-permanent dialogue communication sessions consist of requests and responses that occur

22:20 between applications session layer services are commonly used in application environments that make use of remote procedure calls

22:29 now the services provided by session layer are number one authentication number two

22:36 authorization and number three session restoration the session layer of the USA model is responsible for session

22:45 checkpointing and Recovery it allows information of different streams perhaps originating from different sources to be properly

22:53 combined or synchronized an example usage of the session layer is session beans which are

23:00 only active as long as the session is active and are deleted when the session is disconnected developers can use them

23:07 to store information about the user during a web session an example application is web conferencing in which

23:14 the streams of audio and video must be synchronous to avoid so-called lip sync problems flow control ensures that the

23:22 person displayed on screen is the current Speaker another application is in live TV programs where streams of audio and video need to be seamlessly

23:31 merged and transitioned from one to the other to avoid silent air time or excessive

23:38 overlap next is presentation layer the primary goal of this layer is to take care of the syntax and semantics of the

23:46 information exchanged between two communication systems presentation layer takes care that the data is sent in such a way that the receiver will understand

23:54 the information and will be able to use the data language syntax can be different of the two communicating systems under this condition

24:03 presentation layer plays a role translator the various functions of presentation layer are number one translation before being transmitted

24:12 information in the form of characters and numbers should be changed to bit frames the presentation layer is responsible for interoperability between

24:20 encoding methods as different computers use different encoding methods it translates data between the formats the network requires and then the data

24:29 according to the computer number two encryption the presentation layer carries out encryption at the transmitter and decryption at the

24:37 receiver and number three is compression it carries out data compression to reduce the bandwidth of the data to be

24:45 transmitted the primary role of data compression is to reduce the number of bits to be transmitted it is important

24:52 in transmitting multimedia such as audio video text Etc last but not the least is application

24:59 layer it is the topmost layer of OSI model manipulation of data or information in various ways is done in

25:07 this layer which enables user or softwares to get access to the network some Services provided by this layer

25:14 includes email transferring of files Distributing the results to the user directory Services network resources at

25:21 stra various functions provided by application layer are number one mail services this layer provides the basis

25:29 for email forwarding and storage number two Network virtual terminal it allows a user to log onto a remote host the

25:37 application creates software emulation of a terminal at the remote host users computer talks to the software terminal which in turn talks to the host and vice

25:46 versa then the remote host believes it is communicating with one of its own Terminals and allows users to log on

25:54 number three directory services this layer provides access for Global Information about various services and

26:01 26 minutes, 1 second number four is file transfer access and management it is a standard mechanism to access files and manage it users can

26:09 access files in a remote computer and manage it they can also retrieve files from a remote computer next is the TCP

26:16 IP reference model the TCP IP reference model is the network model used in the current Internet architecture the reference model was named after two of

26:24 its main protocols TCP transmission control protocol and IP that is Internet Protocol the developers choose to build

26:33 a pack switch network based on a connectionless inter Network layer the figure shows various layers of TCP IP reference model let us now briefly

26:41 discuss about the various layers of TCP IP model number one the host to network layer that is physical layer the TCP IP reference model does

26:51 not specify in any great detail the operation of this layer except that the host has to connect to the network using some protocol so it can send IP packets

27:00 over it number two the network layer the job of the network layer is to inject
Packets into any network and have them

27:08 travel independently to the destination the layer defines IP for its official packet
format and protocol

27:16 packet routing is a major job of this protocol number three the transport layer the
transport layer is the

27:24 interface between the application layer and the complex Hardware of the Network
it is designed to allow peer

27:31 entities of the source and destination host to carry on conversations data may be
user data or Control Data two modes are available

27:40 full duplex and half duplex in full duplex operation both sides can transmit and
receive data simultaneously whereas

27:48 in half duplex a site can only send or receive at one time any program running in
the

27:55 application layer has the ability to send a message using TCP or UDP which are the
two protocols defined for the transport

28:04 layer the application can communicate with the TCP or the UDP service whichever
it

28:12 requires both the TCP and UDP communicate with the Internet Protocol in the
internet

28:20 layer in all cases communication is a two-way process the applications can

28:26 read and write to the trans transport layer number four the application

28:33 layer the original TCP IP specification described a number of different applications
that fit into the top layer of the protocol

28:42 stack these applications include tnet FTP SMTP and

28:49 DNS tnet is a program that supports the tenet protocol over TCP tenet is a general
two-way

28:57 communication protocol that can be used to connect to another host and run
applications on that host

29:05 remotely FTP that is file transfer protocol is a protocol that was originally designed
to promote the sharing of files among computer

29:14 users it Shields the user from the variations of file storage on different
architectures and allow for a reliable

29:21 and efficient transfer of data SMTP that is simple mail transfer protocol is the
protocol used to

29:30 Transport electronic mail from one computer to another through a series of other
computers along the

29:37 road DNS domain name system resolves the numerical address of a network node
into its textual name or vice

29:45 versa it would translate www.yahoo.com to 204.
29:52 71177 71 to allow the routing protocols to find the host that the the packet is designed
30:00 for now what are the merits of tcpip number one it operates independently number two it is
30:09 scalable number three client server architecture number four Sports a number of routing
30:16 protocols and number five can be used to establish a connection between two computers next is Dem merits of
30:26 tcpip number number one in this the transport layer does not guarantee delivery of packets number two the model cannot be
30:35 used in any other application number three replacing protocols is not easy and number four it
30:43 has not clearly separated its services interfaces and protocols so with this our today's
30:49 lecture is over concluding this today we learned about the need of protocols and standards also we learned about about
30:58 various protocols at different layers of protocol stack hope you understood today's lecture in the next lecture we'll learn
31:05 about few more concepts of networks and cyber security till then goodbye thank you have a nice day
31:13 [Music]
31:28 n
31:34 [Music]